Name : ahoon shokeen 7a Date : irrelevant, local dates per task

RQ: To what extent do rising global temperatures affect Antarctican scientific research from 1980–2100? (not alarmist)

n+3:

climate change — unit summative Action plan

serial number	guiding question/tasks	date and time assign	date and time compl	status
n	task: make action plan	2026-04-19T05:30:50Z	2026-04-19T06:00:32Z	✓
n+1	task: make research journal template	2026-04-21T13:24:40Z	2026-04-21T13:30:[illegible]	✓
n+2	task: create ~~first~~ GQ ~~guiding questions~~	2026-04-21T14:00:00Z	2026-04-21T14:32:07Z	X
n+3	create and answer GQ n: "What do the antarctic researchers think about climate change?"	2026-04-22T11:30:02Z	2026-04-22T ~~12:00:52Z~~ 13:02:57Z	✓
n+4	answer GQ n+1: "How does climate change affect the quality of science done in antarctica?"	2026-04-22T15:08:47Z not required. post-summative annotation; these are NOT nanoseconds	2026-04-22T16:30:37Z	✓
n+5	Fix remotely research journal ~~answer GQ n+2~~	2026-04-23T07:47:13Z	2026-04-23T07:55:00Z	✓
n+6 Studying	answer GQ n+2: "Why is Antarctic climate change so important?"	2026-04-23T13:36:50Z	2026-04-23T14:24:48Z	✓
n+7	answer GQ n+3: "What makes saving Antarctica so important, and why should we do it?"	2026-04-23T14:37:37Z	2026-04-23T15:18:37Z	✓
n+8	rework 4-point solution	2026-04-23T15:38:01Z	2026-04-23T19:56:57Z	✓

Name :

Date :

your entire works is presented with numbers and units which are not relatable.
Some of them are also incorrect.
Speech was shown to a DP physics teacher.
The model for solution seems flawed and defies the fundamental principles of physics.

Why would you use a numbering system or an unit which is broken down to its lowest possible fraction
look at Ci) criteria
Read I & S criterias and follow them.

Ai) 5
Aii) 5
Bii) 6
Biii) 6
Ci) 2
Dii) 2

A 27/4

Name: aboon shokeen 7a

Date: multiple dates, 2026-04

AIP → american institute of physics 4th edition

climate change — unit $n+3$: summative research journal

serial number	guiding question	citation(s) (AIP style)	Research summary
n	What do the Antarctic scientists think about climate change?	[n] M.J. Siegert, et al. "Antarctic extreme events", Frontiers in Environmental Science, 11, (2023). [n+1] E.A. Laws, editor, *Environmental Toxicology* (Springer, 2013).	Antarctic effects on climate change include localized extreme weather such as extreme winds, drifting sea ice, and collapsing ice shelves.[n] A recognizable issue was the O_3 layer depletion in the stratosphere,[n] letting UV rays through. These could harm the researchers if biochemical reaces occured.[n+1] Modelled by:[n+1] $\underbrace{E_{biol}}_{\text{biologically effective irradiance}} = \int_{280\,nm}^{400\,nm} \underbrace{E_\lambda}_{\text{irradiance}} \overbrace{\sigma_\lambda}^{\text{biological weighting function}} d\lambda$
$n+1$	How does climate change affect the quality of science done in Antarctica?	[n+2] A. Graiff et al. "Big data in Antarctic sciences — current status, gaps, and future perspectives", Polarforschung 91, 45–57 (2023).	In Antarctica, it is already hard to conduct research due to the conditions there: low temps, long nights, sea ice, etc.[n+2] All of these make it harder to study the already sparse ecosystems, both macroscopic and microscopic.[n+2] Climate change accelerates this and ruins the research.

Name :_______________ Date :_______________

serial number	Guiding question	citation(s) (AIP style)	Research
$n+2$	Why is Antarctic climate change so important?	[n+3] P. Convey, et al, "Antarctic climate change and the environment", Antarctic Science 21(6), 541–563 (2009). [n+4] S. R. Rintoul, et al, "Chapter 4.6: The antarctic circumpolar current system", in International Geophysics, (2001), pp. 271–30	The Antarctic system is closely coupled with the global climate system influencing various parts.[n+3] A wind system creates a current which is called the Antarctic Circumpolar Current (ACC).[n+3] This connects the oceans of the Earth.[n+4] 30% of the CO_2 created by humans is sequestered by the sea,[n+3] being a natural carbon sink. It is also the world's most biologically productive ocean.[n+5]
$n+3$	What makes saving Antarctica important, and why should we do it?	[n+5] D. K. A. Barnes et al, "Societal importance of Antarctic negative feedbacks on climate change: blue carbon gains from sea ice, ice shelf and glacier losses", Die Naturwissenschaften 108(5) 43 (2021).	There are 3 types of ice: sea ice, which is a seasonal component of marine ice; fast ice, shore-attached sea ice; and brash ice from ice shelf collapses. These make up 2.5 Mm^2 – 20 Mm^2 of the ice on the planet.[n+5] This ice stabilizes water column, provides a comfortable environment for species like krill, and reduces gas and heat flux.[n+5] It also harbours interesting life, such as bacteria.[n+4]

Name: aroon shaheen 7a   Date: 2026-04-23

climate change — unit n+3: 4-point solution (summative) — reverted

[1] Short-term:

1.1 Helio-centric-orbit reflectors to reflect some of the sun's heat

1.1.1 a novel geoengineering (GE) project that involves reducing incoming insolation with parasols or sunshades. A design proposed ~~to~~ involves a $7 \cdot 10^5$ km² reflective surface located in a non-Keplerian orbit (orbits that take into account more than one gravitational perturber) near L_1.[n+6] This serves 2 purposes:

1.1.1.1 Reduce insolation by about 3.4 $W \cdot m^{-2}$ to lower the avg. temperature.

1.1.1.2 Produce electricity to beam to earth (renewable energy!)

1.2 Ocean phytoplankton fertilization and sequestration

1.2.1 The goal is to fertilize nitrogen-rich chlorophyll-low regions of the sea,[n+7] reversing the 286 Pg of emitted C. For this to be effective, 160 Pg must be sequestered by 2100. When atmospheric CO_2 enters the ocean, it is converted from being an organic compound to 3 inorganics: (aqueous CO_2, bicarbonate HCO_3, and $CO_3^{=}$) ions, of which HCO_3 & $CO_3^{=}$ have a negative charge. This is the natural cycle.[n+7]

1.2.2 Model simulations conducted in 1991 and 1993[n+7] simulated 10–15% of the global ocean being fertilized.[n+7] The results demonstrated the viability of the idea.

Name : ________________ Date : ________________

2 Long-term:

2.1 Ice sheet conservation

2.1.1 buoyantly tensioned flexible barriers that block warm water from flowing into the ice sheets. This shall not do any tethering, rather, they shall be held up with their buoyancy.[n+8] The panels could bend and slide, making them more flexible and less prone to breaking.

2.2 A Dyson swarm

2.2.1 The Dyson sphere was introduced by Freeman J. Dyson in 1960. He suggested that a planet's material could be used to surround a ~~sun~~ star — for our solar system, the mass of that material would be about 2 Qg.[n+9] It would be 2–3 m thick, and would surround the inner planets.[n+9] The issue was that this would be structurally liable to collapse. He then proposed a swarm of small panels.[n+9] This could work, and this would eliminate the need for any fossil fuels. The annual power consumption of the Earth in 2019 was 18.5 TW. The sun has a power output of 320 YW. Per second.[n+9] Clean, renewable (until the sun dies), and powerful.

Citations:

n+6 V. Carestone, and L. Johnson, "Mitigating climate change with Earth orbital sunshades", NASA

n+7 M. Ecol, and P. Ser, "Marine Ecology Progress Series", Marine Ecology Progress Series, (2008)

n+8 B. Keefer et al, "Feasibility of ice sheet conservation using seabed anchored curtains", PNAS Nexus 2(3), (20[illegible])

n+9 J. Smith, "Review and viability of a dyson swarm as a form of dyson sphere", Physica Scripta 97 (12), 122001 (2022).

Name : ahoon shokeen 7a

Date : 2026 - 04 - 24

climate change — unit n+3: summative essay

As an antarctican scientist, I believe (and have evidence for it) that the climate is changing, getting warmer. Although an increase of 1 nK·s^{-1} ?! may not seem like a lot, it adds up. After 1Gs, or 32 years, the temperature would have gone up by a whole kelvin. This is even worse in the polar regions, like the arctic and antarctic. In Antarctica, climate change — according to M.J. Siegert, if you only trust peer-reviewed papers — has many effects. One is the extreme winds. Normally in Antarctica, we get wind gusts of about 28 m·s^{-1}. Thanks to climate change, they are getting stronger. Collapsing ice sheets are a major problem as well, because they undermine our safety and the natural landscape. Which does look really nice. Remember when aerosols dissolved the stratospheric O_3 from the south pole? The sun's UV radiation was let in, dangerously increasing high energy photons ?. Even after it was closed, we still feel the aftereffects. It's also a lot harder to conduct experiments: it was hard previously, but even more so now. The days and nights are each 15.5 Ms long, with dangerously low temperatures of 223 K (one of the Russian centres). The macroscopic ecosystem is also sparse, so we devote more of our time to the microscopics. To do this, we need proper equipment, which needs to withstand the Antarctic desert. They are expensive. And this is not just me rambling about my problems, these actually affect the world. Our sea is coupled with the global climate, due to it being a carbon sink for 30% of sequestered carbon. All of the oceans are, according to S. Rintoul, connected via a current system called the Antarctic Circumpolar Current.

Name :
Date :

Fundamental errors in your concept.

Now, let us talk about solutions. The following solutions have been analysed with others' data on the same. As a quick fix, we can launch reflective panels — with an area of $7 \cdot 10^5 \ km^2$ or so — and put into orbit around L_1. L_1 is where the sun's gravity and earth's gravity cancel each other out. It is not as stable as L_4 or L_5, but it has one major asset. L_1, due to its "neutrality", orbits the sun at the same speed as the earth. The goal here are to reduce the incoming solar radiation by $3.4 \ W \cdot m^{-2}$, and of course, indulge in a little profit from the sun's powerful light as electricity. It is ambitious, but it is not impossible. It's necessary. However, we should look to the future. In the long term, if that panel fails or blocks too much sunlight, our world could become a good setting for a hard-science-fiction novel. Because of this glaring issue, we turn to a better solution: the Dyson swarm. It was proposed originally as a sphere, not a swarm, but the core idea is to surround the inner planets with a swarm of solar panels, harvesting as much energy as they can from the sun, eliminating fossil fuels. Who needs to burn stuff when you have a free nuclear fusion reactor?

Essentially, we should work towards fixing climate change now. These solutions may seem esoteric, but they are possible, and it is all just a matter of funding research. We must convince people and governments that this issue is easy to fix.